JAVIER REBULL, M.Sc.

Senior Full Stack Engineer | AI & ML Specialist | Software Architect

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Summary

Senior Full Stack Engineer and AI/ML Specialist with 20 years in banking technology and enterprise software engineering across financial services, payments, and credit cards. Experienced in designing and leading mission-critical platforms, including an IBM Mainframe-to-cloud migration with zero critical incidents. Hands-on experience deploying ML models in production with MLOps and building cloud-enabled enterprise applications, from banking platforms to EpiForecast-MX, a national-scale AI forecasting platform.

Technical Skills

Programming Languages: Python | Java | JavaScript | SQL | COBOL | SAS
AI / Machine Learning: Deep Learning | NLP | Time Series Forecasting | Computer Vision | Generative AI & LLMs
Frameworks & Libraries: PyTorch | Scikit-learn | XGBoost | Pandas | NumPy | Prophet | DeepAR | React
Cloud Technologies & DevOps: AWS (SageMaker, Lambda, S3, EC2) | Scalable Cloud Architecture | Mainframe-to-Cloud Migration | MLOps | CI/CD | Docker | Linux | Git/GitHub
Software Engineering: Full Stack Web Applications | System Architecture | Microservices & Distributed Systems | REST/SOAP APIs | Agile/Scrum Delivery
Databases: Oracle | DB2 | MySQL | MongoDB
Financial Technology: Core Banking | Payments | Credit Cards | Loans | Regulatory (FDIC 370) | High Availability

Work Experience

Senior Associate, Application Development
Santander Bank US — Quincy, MA — Sep 2019 – Present

Promoted from Associate, Application Development (Sep 2019 – Feb 2024)

- Led the Gravity Middleware Project end to end, migrating IBM Mainframe/DB2 to private cloud (Microfocus/Oracle) with zero critical incidents, and modernized 500+ WAS/WebSphere servers into scalable, reliable cloud infrastructure.
- Built end-to-end MLOps pipelines for automated model deployment and monitoring in production banking environments.
- Directed full stack development of OpenBank Certificates of Deposit, from dynamic web UIs to enterprise back ends.
- Delivered the SBNA Loans COVID-19 Relief Program and Regulatory FDIC 370 on time with business and risk teams.
- Designed UI/UX for internal banking applications, raising user satisfaction by 35%.
- Kept distributed banking systems incident-free through root-cause analysis across hardware, network, and service layers, improving uptime and performance, and led design and code reviews on testability and code quality.

Computer Systems Analyst
Vass IT US — Cambridge, MA — Apr 2013 – Sep 2019

- Built Computer Vision solutions for document processing and fraud detection, integrating ML models into client systems.
- Led full stack development of loan-management applications and managed IT project portfolios and PMO activities.
- Owned software products from requirements through deployment and maintenance for client business goals.

Spec Analytics Analyst 2
Citibank Banamex — Mexico City, Mexico — Jun 2010 – Apr 2013

- Designed and optimized secure distributed systems for global financial services, processing millions of daily transactions with high availability.
- Built data-driven algorithms (SAS/Python) that raised marketing response rates 20% and cut delinquency 10%.
- Built interactive dashboards and front-end interfaces for executive reporting and real-time data visualization.

Senior Programmer
Accenture — Mexico City, Mexico — Aug 2007 – May 2010

- Developed and maintained large-scale core financial systems (COBOL, Java) for client banks BBVA Bancomer and Santander México, with zero critical incidents.
- Designed front-end user interfaces for mainframe applications, improving operational efficiency and developer experience.

Systems Analyst & IT Help Desk Analyst
Microsistemas / Ford Credit Company — Mexico City, Mexico Aug 2006 – Aug 2007

- Help desk consultant for Ford's department: system stability, security, and software and hardware support.

Education

Universidad Autónoma de Ciudad Juárez (UACJ)
M.S. in Artificial Intelligence and Data Analytics | Expected 2027
Ciudad Juárez, Mexico
Focus: Smart Optimization, Linear Optimization, Non-Linear Optimization, Data Engineering

Tecnológico de Monterrey (ITESM)
M.Sc. in Applied Artificial Intelligence | Conferred 2026
Mexico City, Mexico
GPA: 4.0/4.0 • Coursework: Data Science, Machine Learning, AI, Data Visualization, Fintech & IoT

The National Polytechnic Institute of Mexico (ESCOM IPN)
B.S. in Computer Science & Electronics Engineering | 2006
Mexico City, Mexico
GPA: 3.28/4.0 • Coursework: Programming, Software Engineering, AI, Networks, Data Structures, Algorithms & OS

AI / ML Projects

- **EpiForecast-MX — National-Scale Epidemiological Forecasting Platform** (2025–2026): Built at Tecnológico de Monterrey in collaboration with the Instituto Mexicano del Seguro Social (IMSS), it forecasts weekly notified cases across Mexico's 32 federal entities on a 52-week horizon and runs in production at epiforecast.mx. Selects per series among Prophet, DeepAR, Ensemble and Stacking models through an auditable rule; M.Sc. capstone.
- **Text Classification with RNNs — AG NEWS Dataset** (Mar 2025): Built an RNN in PyTorch to classify AG NEWS articles, covering data preprocessing, model architecture design, training pipelines, and evaluation.
- **IoT for Sustainable Urban Development** (Feb 2025): Investigated IoT solutions for traffic congestion in Boston and public lighting efficiency in Mexico City using distributed sensor networks.
- **Multilayer Fully Connected Network in NumPy** (Jan 2025): Implemented a neural network from scratch for ASL letter recognition, including forward/backward propagation and performance benchmarking.
- **Stock Price Prediction for Apple** (Nov 2024): Developed and compared LSTM, ARIMA, Prophet, and Naïve predictive models on historical market data using advanced statistical methods.

Conferences & Publications

- **Two papers** — MICAI 2026, Chihuahua, Mexico (Nov 2–6, 2026), Springer, to appear; first author of both: *"Forecasting Weekly Depression Incidence in Mexico: A Multi-Model Framework with Auditable Per-Series Model Selection"* (LNCS, oral; EpiForecast-MX, Tecnológico de Monterrey & IMSS) and *"A Two-Condition Diagnostic for Decoder-Based Multi-Objective Search: When Blind Sampling Beats a Mismatched NSGA-II"* (CCIS, poster; UACJ).
- **Oral presentation** — CALASS 2026, Congrès de l'ALASS, Université de Montréal, Canada (Aug 27–29, 2026): *"De la surveillance épidémiologique à l'intelligence prédictive : le rôle de l'intelligence artificielle dans l'organisation prospective des systèmes publics de santé."* Presented EpiForecast-MX (Tecnológico de Monterrey & IMSS).
- **ePoster** — 26th WPA World Congress of Psychiatry, Stockholm, Sweden (Sep 23–26, 2026), Digital Psychiatry track: *"Population-level prediction of depression, Alzheimer's disease, and Parkinson's disease incidence using artificial intelligence: implications for health service planning."*
- **Poster** — 2nd Public Health Conference & 2nd HSG Europe Preconference, Skopje, North Macedonia (May 21–22, 2026): *"AI-Based Forecasting to Shift Epidemiological Surveillance toward Predictive Approaches: Depression, Alzheimer's, and Parkinson's as Use Cases."*
- **Article** — Medium: *Forecasting Dengue in Mexico: An End-to-End Data Science Story.*

Certificates: rebull.org/constancias

Training & Certifications

Python Essentials 1 & 2 — Cisco (Jul 2026) • Lifelong Learning — Certiprof (Jan 2026) • AWS Certified AI Practitioner (Dec 2025) • Scrum Master Professional — SMPC® (Jun 2025) • Generative AI in Action — IBM SkillsBuild (Jun 2025) • AWS Certified Cloud Practitioner (Jan 2025)